

PIONEER JUNIOR COLLEGE

JC2 PRELIMINARY EXAMINATION
HIGHER 2

CANDIDATE
NAME

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CT
GROUP

1	4			
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INDEX
NUMBER

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CHEMISTRY

9647/02

Paper 2 Structured

15 September 2015

2 hours

Additional Materials: Data Booklet

READ THESE INSTRUCTIONS FIRST

Write your CT group, index number and name on all work you hand in.
Write in dark blue or black pen on both sides of the paper.
You may use a soft pencil for any diagrams, graphs or rough workings.
Do not use staples, paper clips, highlighters, glue or correction fluid.

Answer **all** questions.

At the end of the examination, fasten all your work securely together.
The number of marks is given in brackets [] at the end of each question or part question.

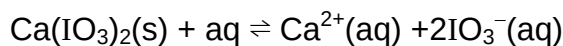
FOR EXAMINER'S USE			
Paper 2			
1	/ 12	5	/ 15
2	/ 7	6	/ 12
3	/ 11	Penalty	sf units
4	/ 15	Total	/ 72

Answer **all** the questions in the spaces provided.

1 Planning (P)

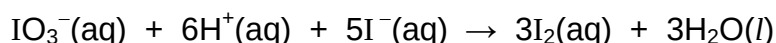
- (a) The solubility of calcium iodate(V), $\text{Ca}(\text{IO}_3)_2$, is approximately 5 g dm^{-3} .

When solid calcium iodate(V) is added to water, a small amount dissolves to form a saturated solution, establishing an equilibrium between the solid and its aqueous ions.

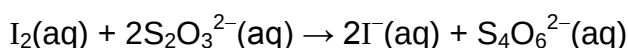


You are given a solid, **FA1**, containing $\text{Ca}(\text{IO}_3)_2$. You are asked to determine the exact concentration of IO_3^{-} ions in a saturated solution of $\text{Ca}(\text{IO}_3)_2$, and hence the solubility product of $\text{Ca}(\text{IO}_3)_2$, by carrying out two reactions:

- reaction of acidified IO_3^{-} with I^{-} to liberate aqueous I_2 ,



- titration of the I_2 liberated against $\text{Na}_2\text{S}_2\text{O}_3$.



- (i) Using the information provided, you are required to write a plan to determine, as accurately as possible, the solubility product of calcium iodate(V).

You are provided with the following:

- **FA1**: solid calcium iodate(V), $\text{Ca}(\text{IO}_3)_2$
- **FA2**: aqueous potassium iodide, KI
- **FA3**: aqueous sodium thiosulfate, $\text{Na}_2\text{S}_2\text{O}_3$
- **FA4**: aqueous hydrochloric acid, HCl
- starch indicator
- apparatus normally found in a college laboratory

Your plan should include:

- brief, but specific details of the apparatus you would use, bearing in mind the levels of precision they offer,
- details, including quantities for the preparation of the saturated solution of $\text{Ca}(\text{IO}_3)_2$ from solid **FA1**,
- essential details of the titration procedure.

- (ii) A student found that 25.0 cm^3 of the saturated solution of $\text{Ca}(\text{IO}_3)_2$, upon treatment with excess **FA2** and **FA4**, requires $v \text{ dm}^3$ of $M \text{ mol dm}^{-3}$ **FA3** for complete reaction.

Show how you would use her results to determine the solubility product of $\text{Ca}(\text{IO}_3)_2$ in terms of v and M .

[3]

- (b) **FA5** is a solution containing a mixture of bromide and iodide ions.

You are provided with the following reagents and laboratory apparatus:

- $\text{HCl}(\text{aq})$
- concentrated $\text{NH}_3(\text{aq})$
- dilute $\text{NH}_3(\text{aq})$
- $\text{AgNO}_3(\text{aq})$
- $\text{HNO}_3(\text{aq})$
- deionised water
- test tubes
- filter funnel
- filter paper

Using **only the reagents and materials provided**, devise a sequence of steps by which the two anions could be separately obtained as **precipitates**.

In your plan, you should:

- give a full description of your proposed sequence of steps, numbering each step;
- state the location of each anion (e.g. in a solution or precipitate) after each step.

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- 2 Stink bombs can be constructed using solid ammonium hydrosulfide, which decomposes to form ammonia and hydrogen sulfide readily.



- (a) Calculate the average M_r of the gaseous mixture.

[1]

- (b) When 0.500 g of solid NH_4SH was placed in a 350 cm^3 sealed flask at 20°C and left to equilibrate for a day, 0.265 g of solid was found to remain.

Using your result from (a), calculate the total pressure of gas in the sealed flask.

[2]

- (c) (i) Write an expression for K_p of the decomposition of ammonium hydrosulfide.

[1]

- (ii) Using your answer in (b), determine the K_p for the decomposition of ammonium hydrosulfide.

[2]

- (iii) Explain how the position of equilibrium would be affected when more solid NH_4SH is added to the equilibrium mixture.

.....
[1]

[Total: 7]

3 (a) The solubility of Mg(OH)_2 in water at 25°C is $1.20 \times 10^{-2} \text{ g dm}^{-3}$.

(i) Calculate the value of K_{sp} of Mg(OH)_2 , stating its units.

[2]

(ii) Predict whether a precipitate of Mg(OH)_2 would be observed, when equal volumes of a $0.10 \text{ mol dm}^{-3} \text{ Mg}^{2+}(\text{aq})$ solution and a buffer solution at pH 9.5 are mixed.

[2]

(b) Mg(OH)_2 undergoes decomposition on heating to give the corresponding oxide. State and explain how the decomposition temperature of Ca(OH)_2 would compare with that of Mg(OH)_2 .

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.....[2]

- (c) (i) Construct an energy level diagram using the following data, together with relevant data from the *Data Booklet* to calculate the lattice energy of magnesium oxide.

enthalpy change of atomisation of Mg(s)	+150 kJ mol ⁻¹
first electron affinity of oxygen	-147 kJ mol ⁻¹
second electron affinity of oxygen	+753 kJ mol ⁻¹
enthalpy change of formation of MgO(s)	-602 kJ mol ⁻¹

Energy



[3]

- (ii) Suggest, with reasons, how the magnitude of the lattice energy of MgF₂(s) might compare to that of MgO(s).

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.....[2]

[Total: 11]

- 4 Copper is a typical *transition element*. It has more than one oxidation state and many of its compounds have colours in the blue-green-yellow part of the visible spectrum.

(a) What do you understand by the term *transition element*?

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.....[1]

(b) Explain why the first ionisation energies of transition elements, from titanium to copper, are relatively invariant.

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.....[2]

(c) What feature of transition elements, such as copper, allows them to show variable oxidation states?

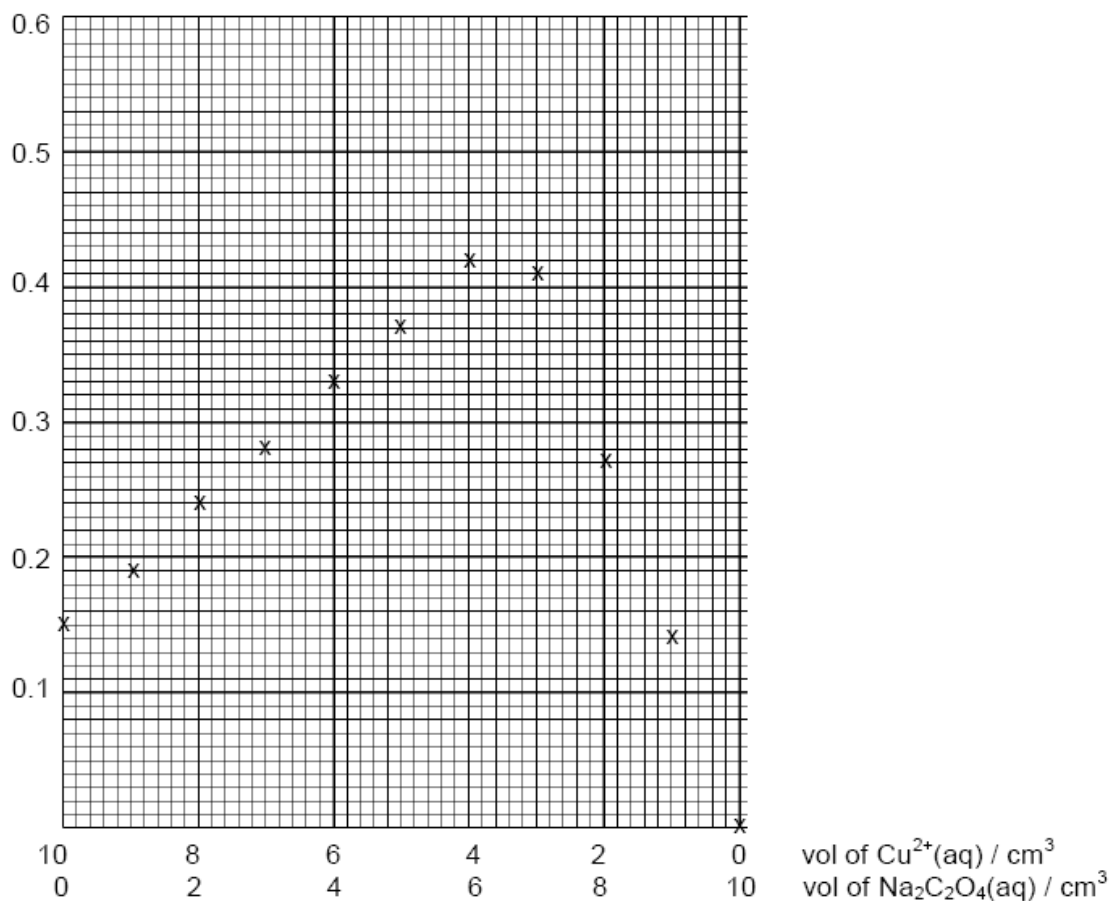
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.....[2]

- (d) Copper forms many complexes with ligands such as H_2O , NH_3 and Cl^- .

To determine the stoichiometry of a copper(II) ethanedioate complex, the colour intensities of solutions containing different proportions of the ligand, $\text{C}_2\text{O}_4^{2-}$, and Cu^{2+} ion were measured.

The intensity of the colour depends on the concentration of the complex ion. The higher the concentration of the complex, the higher is the colour intensity. In one such experiment, the results obtained for different mixtures of $1.0 \times 10^{-4} \text{ mol dm}^{-3} \text{ Cu}^{2+}(\text{aq})$ and $1.0 \times 10^{-4} \text{ mol dm}^{-3} \text{ Na}_2\text{C}_2\text{O}_4(\text{aq})$ were plotted below.

relative colour intensity



- (i) By drawing two lines of best fit through the plotted points, determine the mole ratio of $\text{Cu}^{2+}(\text{aq})$ to $\text{C}_2\text{O}_4^{2-}(\text{aq})$ in the complex?

[2]

- (ii) The copper(II) ethanedioate complex has an octahedral shape and it retains some water molecules as ligands.

Suggest a formula for the complex, including its overall charge.

.....[1]

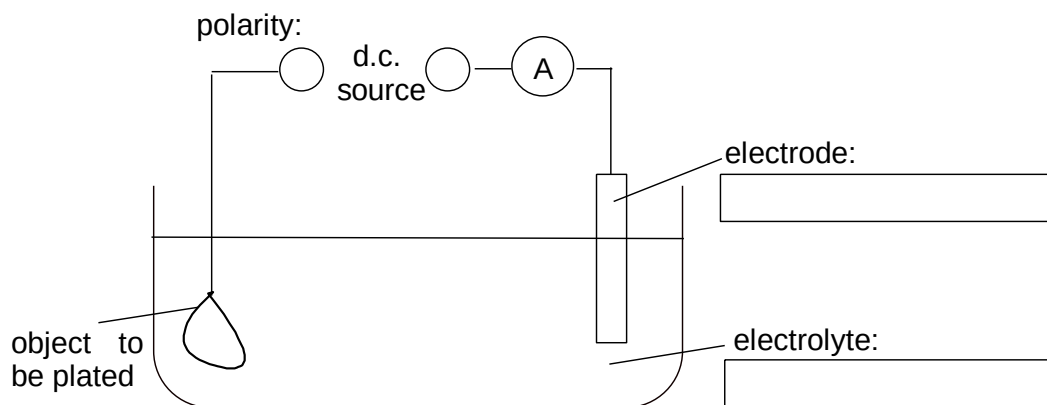
- (e) Explain why complexes of Cu(II) are coloured whereas complexes of Cu(I) are colourless.

.....

[3]

- (f) Because of its increased scarcity, cheaper copper ornaments are no longer made from the solid metal, but from iron that has been copper plated.

- (i) Complete the following diagram showing the set-up for a copper electroplating process. Show clearly the polarity (+/-) of the power source and suggest a suitable electrode and electrolyte to be used.



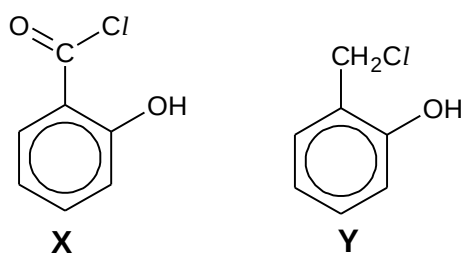
[2]

- (ii) A current of 0.600 A is passed through the electroplating cell. Calculate the time required, in seconds, to deposit a mass of 0.600 g of copper onto the ornament.

[2]

[Total: 15]

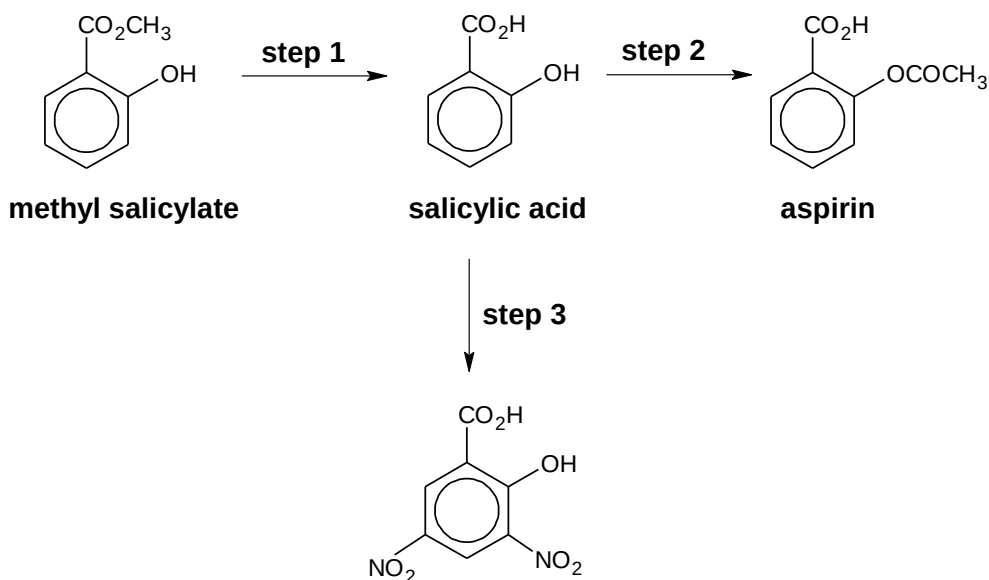
- 5 (a) Describe a chemical test which can be used to distinguish between compounds X and Y. State what would be seen with each compound.



.....

[2]

- (b) Salicylic acid is an important precursor in the production of aspirin, a widely used anti-inflammatory medication. Salicylic acid is commonly prepared from methyl salicylate (oil of wintergreen). The general production of aspirin is outlined below together with some reactions of salicylic acid.



- (i) Give the reagents and conditions for **steps 1, 2 and 3**.

step 1:

step 2:

step 3:

[3]

- (ii) Write an equation for the reaction of salicylic acid with sodium carbonate.

[1]

- (iii) State and explain how the acidity of methyl salicylate compares with that of aspirin.

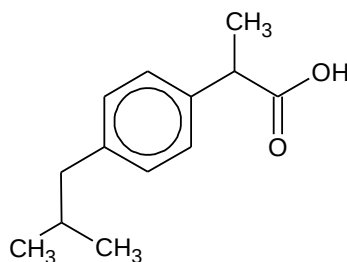
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.....[3]

- (iv) 4-hydroxybenzoic acid, an isomer of salicylic acid, has a melting point of 215 °C while salicylic acid has a melting point of 159 °C.

Suggest an explanation for the difference in the melting points of the two compounds.

.....
.....
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.....[2]

- (c) Ibuprofen has similar anti-inflammatory drug properties as aspirin. A *racemic mixture of ibuprofen* is produced industrially.



Ibuprofen

- (i) With the aid of suitable diagrams, explain what is meant by the term *racemic mixture of ibuprofen*.

[2]

- (ii) It was determined that only one enantiomer, the (+)-ibuprofen has anti-inflammatory properties.

An optically pure enantiomer of (+)-ibuprofen has an optical rotation of $+54.5^\circ$. When a racemic mixture of ibuprofen is consumed, an enzyme in the body is able to convert the biologically useless (-)-ibuprofen to (+)-ibuprofen.

A patient consumes a racemic mixture of ibuprofen and a sample of his blood was taken. The ibuprofen in his blood was found to have an optical rotation of $+32.7^\circ$. Determine the percentage of (-)-ibuprofen that was converted to (+)-ibuprofen by the enzymes in the patient's body.

[2]

[Total: 15]

- 6 (a) Suggest, with reasons, how the basicity of phenoxide ion, $\text{C}_6\text{H}_5\text{O}^-$ and ethoxide, $\text{CH}_3\text{CH}_2\text{O}^-$ might compare with that of hydroxide ion, OH^- .

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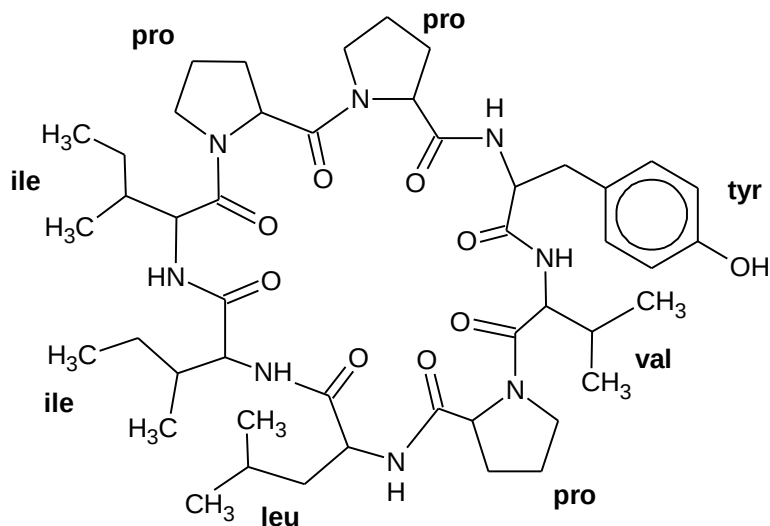
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.....[3]

- (b) Hymenistatin is a cyclic peptide isolated from sea sponges. It is found to have immunosuppressive effects. The structure of hymenistatin is given below along with the abbreviated names of its constituent amino acids: **pro**, **leu** etc.



- (i) The enzyme chymotrypsin hydrolyses proteins or polypeptides at the carboxylic end of the amino acid tyrosine, **tyr**.

Use the abbreviated names of the amino acids to state the primary structure of hymenistatin after its hydrolysis by chymotrypsin.

.....[1]

- (ii) Muscle tissues found in meat has a β -pleated sheet structure.

Draw a diagram showing **two** strands of a β -pleated sheet structure using the formula below. Indicate clearly the interaction between the two strands and show how the R groups are directed.



[2]

- (iii) Suggest why the β -pleated sheet structure is unlikely to be found in hymenistatin.

.....

.....[1]

- (iv) The biological function of hymenistatin depends on its three-dimensional shape.

Suggest, with a reason, the main type of interaction involved in maintaining the three-dimensional shape of hymenistatin.

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.....[2]

- (v) Describe how hymenistatin can be hydrolysed in the laboratory to form a mixture of its constituent amino acids?

.....

.....[1]

- (vi) Electrophoresis can be used to separate the mixture of amino acids. This technique involves placing the mixture at the centre of an agarose gel placed in an aqueous solution of a suitable pH. An electric potential is then applied through electrodes which are connected to the two ends of the gel strip.

Which pair of amino acids found in hymenistatin is not likely be separated by electrophoresis? Explain your answer.

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.....[2]

[Total: 12]

End of Paper